UNIVERSITY OF GRAZ
Department of Mathematics
and Scientific Computing

MAT.B0204UB MATHEMATICAL MODELLING
SUMMER SEMESTER 2025/26

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EXERCISE SHEET 5

Due: 27 May 2026, 10:00

EXERCISE 5.1 (Epidemiology: the SIRS model)
In the context of epidemiology, we consider the SIRS model below where the unknow S stands for Susceptible, I for Infectious and R for Recoverded,

$$\begin{cases} \dot{S} = -\beta SI + \mu R, \\ \dot{I} = \beta SI - \nu I, \\ \dot{R} = \nu I - \mu R, \end{cases} \tag{1}$$

with $\beta, \mu, \nu > 0$ and where the total population $N = S + I + R$ is constant in time.

(i) Use the total population N to reduce (1) to a 2 dimensional system. Then, identify the steady states and perform the linear stability analysis.

(ii) Perform the global qualitative analysis of the reduced nonlinear system obtained in (i). Namely, analyse the function below,

$$W(x, y) := y - \bar{I} - \bar{I} \ln \frac{y}{\bar{I}} + \frac{1}{2\bar{S}} \left(x + y - \bar{S} - \bar{I} \right)^2,$$

where $(\bar{S}, \bar{I}))$ is the non trivial steady state of the reduced model in (i).

(iii)* (Optional) Assume $\nu = 0$ and study the qualitative analysis of the obtained system, by outlining the main difference with the SIRS model (1).